

Computational Neuroscience: Analysis of Brain Patterns Using Machine Learning Algorithms

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Abstract

Perceptual decisions are usually based on current sensory information, but previous choices may also influence later judgments. This project examines whether repeated perceptual decisions depend only on new evidence or are biased by previous choices. Participants performed a visual task in which they classified sequences of oriented stimuli as cardinal or diagonal across repeated presentations. The design allowed us to test whether decision stability reflected simple response repetition or a biased integration of new sensory evidence.

Behavioral results showed that participants often repeated previous decisions, even when the stimulus order changed. However, they remained sensitive to the current sensory evidence, suggesting that later choices combined new information with previous decisions. Confidence also contributed to this effect, as high-confidence decisions were more likely to be repeated. Computational modeling showed that previous decisions were best explained as influencing evidence accumulation rather than only the initial response bias. Machine learning and EEG analyses provided complementary evidence that decision history and neural activity contained information about later behavior. Overall, these findings suggest that perceptual confirmation bias stabilizes previous interpretations and influences how new sensory evidence is used.

Keywords: perceptual decision-making, confirmation bias, confidence, evidence accumulation models, metacognition, EEG



1 INTRODUCTION TO THE PROJECT

Perceptual decisions are part of everyday life. In many situations, we need to make a decision based on sensory information that is not completely clear, such as identifying an object from a distance or interpreting an ambiguous visual scene. Although these decisions may seem to depend only on the information available at that moment, previous choices can also influence how new information is interpreted.

This project focuses on how previous perceptual decisions affect later choices when similar sensory evidence is presented again. More specifically, it studies perceptual confirmation bias, that is the tendency to maintain a previous interpretation and to process new information in a way that is consistent with that earlier decision.

To study this, participants performed a visual decision-

making task in which they had to classify sequences of oriented stimuli as either cardinal or diagonal. The same sequence or a physically different sequence with similar sensory evidence could be presented more than once. This design made it possible to examine whether later responses were influenced by the current stimulus or if they were driven by the previous decision of the participant.

The project combines behavioral analyses, computational models of evidence accumulation, machine learning methods and electroencephalography (EEG) analyses. Together, these approaches allow the same question to be studied from different perspectives, in other words, study how participants behave, how the decision process can be modeled, how well future responses can be predicted and how sensory information is represented in brain activity.

2 OBJECTIVES

The main objective of this project is to understand how prior perceptual decisions influence later decisions when similar sensory information is presented again. More specifically, the project aims to examine whether decision repeti-

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tion reflects a simple tendency to repeat prior responses or whether previous choices actually bias the way new sensory evidence is processed.

2.1 Specific Objectives

1. To describe how often participants repeat or change their perceptual decisions across repeated stimulus presentations.
2. To examine whether later decisions are still influenced by current sensory evidence or whether they are more strongly influenced by previous responses.
3. To analyze whether confidence in a previous decision increases decision stability and reduces the probability of changing one's mind.
4. To compare different computational mechanisms of confirmation bias using Hierarchical Sequential Sampling Models (HSSM), focusing on whether the bias affects the starting point of the decision process or the way evidence is accumulated.
5. To test whether machine learning models can predict later decisions from current sensory evidence, previous choices, confidence and other behavioral variables.
6. To explore whether EEG activity contains information about stimulus orientation, perceptual decisions and confidence and whether these patterns change across repetitions.

3 CONTEXT AND STATE OF THE ART

Perceptual decision-making is the process through which the brain uses sensory information to choose between different possible responses. In many situations, the information available is not completely clear. For this reason, the brain often needs to combine information over time before making a decision.

This process is often described using Drift Diffusion Models (DDM). These models assume that the human brain gradually accumulates evidence in favor of one option or another until there is enough evidence to make a decision. This framework is useful because it helps explain both the decision that is made and the time needed to make it [1, 2].

For a long time, many perceptual decision-making studies treated each decision as independent from the previous ones. In this view, each response would mainly depend on the sensory evidence available in the current trial. However, more recent studies have shown that this is not always the case. Previous choices can influence later judgments, even when new sensory information is available. This effect is usually described as choice history bias or serial dependence, suggesting that both perception and decision-making are influenced by recent experience [3, 4, 5, 6].

This influence of previous choices can be useful when the environment is stable. If similar information appears again, relying partially on recent experience may help people make decisions more efficiently. However, it can also become a problem when the new information is different

from what was expected. In that case, the previous decision may make people less sensitive to evidence that goes against their first interpretation.

A more specific form of this effect is perceptual confirmation bias. In this case, a previous decision may not only increase the probability of repeating the same response, but may also affect how new evidence is processed. Evidence that is consistent with the previous choice may receive more weight, while evidence that contradicts it may have less influence. This is related to the idea that people tend to maintain consistency with their previous decisions once an initial judgment has been made [7]. As a result, decision stability can sometimes reduce sensitivity to new or contradictory information [8, 9, 10].

An important question is whether repeated decisions are only the result of a tendency to repeat the same response or whether the previous decision actually changes how new sensory information is used. These two possibilities are different. In the first case, the participant would simply be biased to repeat the previous answer. In the second case, the participant would still process the new stimulus, but this processing would be influenced by the previous choice. Distinguishing between these possibilities is important for understanding how confirmation bias operates in perceptual decisions.

Confidence is also important in this context. A decision made with high confidence may be more resistant to change, because the participant is more likely to trust that previous response. For this reason, confidence can help stabilize previous choices across repetitions. This is closely related to metacognition, which refers to the ability to evaluate one's own performance and to judge if a decision is likely to be correct or not [11, 12, 10].

In this project, these ideas are studied using a repeated perceptual decision-making task. This type of task is useful because it makes it possible to test whether later decisions are based only on the current sensory evidence or are also influenced by previous choices. Therefore, confirmation bias can be studied not only as a tendency to repeat responses, but also as a possible change in the way new evidence is used.

4 DATA

The project used behavioral and EEG data from 25 valid participants. Three participants were excluded because the data was incomplete and could not be used consistently during the analyses. Participants performed a visual perceptual decision task. In each trial, they saw a sequence of oriented visual stimuli presented one after another. After the sequence, they had to decide whether the sequence of stimuli was mainly cardinal or diagonal and then report how confident they were in that decision. A key aspect of the experiment was that the visual evidence could be presented up to three times. In some trials, the exact same sequence was repeated, while in others the same orientations were presented again in a different order. This design made it possible to test whether later decisions were influenced only by the current visual information or also by the decision made in the previous presentation. This repetition structure was coded with the variable *nrep*, where *nrep* = 0 corresponded to the first presentation, *nrep* = 1 to the second presenta-

tion and $nrep = 2$ to the third presentation. This structure is shown in Fig. 1.

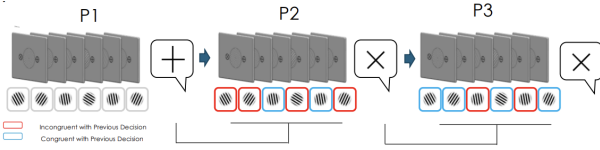


Fig. 1: Schematic representation of the repeated visual decision-making task. Participants viewed a sequence of oriented stimuli and classified it as cardinal or diagonal. Across the three presentations, the same orientations could appear again in the same order or in a different one. After the first presentation, each stimulus could be described as congruent or incongruent depending on whether it supported or contradicted the previous decision of the participant. This allowed the analyses of how previous choices influenced the processing of new sensory evidence.

The behavioral dataset included several variables, such as the decision of the participant, reaction time, confidence rating, accuracy, stimulus condition, sensory evidence and information about previous decisions. The main sensory variable was the decision variable (rDV), which indicated how strongly each stimulus sequence supported a cardinal or a diagonal response. Values close to zero corresponded to ambiguous stimuli, meaning sequences that did not clearly support either a cardinal or a diagonal response. In contrast, more extreme values represented sequences that were easier to classify because they contained more clearly dominant orientations.

EEG was recorded during the task with 32 channels and a sampling rate of 500 Hz. The experiment also included a simpler EEG task called the localizer. In this task, participants saw visual stimuli, but they did not have to integrate information across repetitions as in the main task. This localizer was used to measure basic visual responses to stimulus orientation and to compare them with the activity recorded during the main task.

Before starting the analyses, the data was preprocessed to remove atypical trials. Very fast or very slow responses were excluded because they could reflect distractions or accidental responses rather than normal task performance. This filter was applied separately for each participant, so each person was compared with their own response time distribution. Specifically, only trials with reaction times higher than the participant's mean plus four standard deviations were removed. This affected 63 out of 6600 trials, which is about 1% of the dataset.

5 METHODOLOGY AND DEVELOPMENT

5.1 Behavioral Analyses

The first step was to analyze the behavioral data in order to understand the main patterns in the responses of the participants. These analyses focused on accuracy, reaction times, confidence and the probability of repeating or changing a previous decision across stimulus presentations.

Descriptive analyses were first used to check whether participants performed the task consistently. This was important because, before studying confirmation bias, it was necessary to verify that participants understood the task and

were responding in a stable way. Accuracy and reaction times were therefore examined across repetitions and trial types.

The next step was to study decision repetition. Since the same orientations could be presented again either in the same order or in a different one, the analyses examined how often participants repeated their previous response and whether this depended on the type of repetition. This made it possible to test whether decision stability was simply caused by seeing the exact same sequence again or whether it also appeared when the order of the orientations changed.

Initially, some behavioral effects were explored using simple logistic regressions. However, these models were not ideal because they do not take into account that participants can differ from each other. For example, some participants may be generally more confident, faster, slower or more sensitive to sensory evidence than others. For this reason, the main analyses were performed using Linear Mixed Models (LMMs) and Generalized Linear Mixed Models (GLMMs). These models are useful in this type of experiment because they allow the analysis to estimate general effects while also taking into account individual differences between participants. In other words, they make it possible to study the overall pattern in the data without assuming that all participants behave in exactly the same way.

Psychometric analyses were also performed to examine whether the current decision depended on the orientations presented in the current sequence. For example, when the current sequence contained more diagonal orientations, participants should be more likely to respond diagonal, whereas sequences with more cardinal orientations should lead to more cardinal responses. This was important because it helped test whether participants based their current response on the current visual evidence. If participants were only repeating their previous decision, the orientations in the current sequence would not strongly affect the current response. However, if participants were still using the current stimulus, their responses should change according to the orientations presented in that sequence. These analyses therefore helped distinguish between simple repetition of the previous response and a biased use of current sensory evidence.

Additional behavioral analyses, including reaction times, psychometric curves and confidence analyses, are reported in Appendix A.

5.2 Computational Modeling with HSSM

To study the decision process in more detail, HSSM models were fitted using the DDM framework. These dynamic models describe decisions as a gradual accumulation of evidence until enough evidence is available to make a response. Unlike descriptive behavioral analyses, HSSM models can model both the choice that participants make and the time they take to respond. This helps distinguish between different possible mechanisms involved in the decision-making process.

This was useful because the behavioral analyses showed what participants did, but not exactly how the decision process changed. With HSSM, it was possible to estimate hidden aspects of the decision process, such as the starting point and the drift rate. The starting point indicates whether

the decision process begins closer to one response or the other, while the drift rate reflects how evidence is accumulated over time. These parameters helped test which part of the decision process was influenced by previous choices.

Two possible mechanisms of confirmation bias were compared. In the first model, the previous decision affected the starting point of the next decision. This means that, before processing the new evidence, the participant already started slightly closer to the response chosen before. In the second model, the previous decision affected the drift rate. This means that the new evidence was accumulated in a way that was influenced by the previous choice.

The difference between these two models is important because they represent different explanations of confirmation bias. A starting point bias would suggest that the participant begins the next decision with an initial preference for the previous response. A drift rate bias would suggest that the previous decision changes how the new sensory evidence is processed during the decision itself.

To determine which mechanism explained the data better, the two models were compared by testing how well they could predict the decisions of the participants. This comparison was done using leave-one-out cross-validation (LOO-CV). This means that the model is tested on data that was left out during the training, instead of being evaluated only on the same data used to build it. In this way, the comparison gives a more realistic measure of how well each model can explain new observations.

The comparison was summarized using *elpd_loo*, a standard measure used in LOO-CV. This value reflects the predictive performance of each model, with higher values indicating better prediction of the data. Therefore, the model with the higher *elpd_loo* value was interpreted as having stronger support.

As an additional check, predictive accuracy was also calculated by comparing the decisions predicted by each model with the real decisions made by the participants.

Before applying the models to the real data, simulations and parameter recovery analyses were performed. In the simulations, artificial data was generated with known bias effects. The models were then fitted to these simulated data to test whether they could recover the effects that had been introduced. This step was important because it checked that the models were able to identify the expected bias mechanisms before using them to interpret the real experimental data. The simulations and parameter recovery analyses are reported in Appendix B.

5.3 Machine Learning Analyses

Machine learning analyses were used as a complementary way to test whether participants' behavior could be predicted from the variables available in the task. While the behavioral and computational analyses focused on explaining the mechanisms behind the responses, the machine learning analyses focused on prediction. In other words, they tested whether information from the stimulus, previous choices and confidence was enough to predict what participants would decide later.

Two main targets were studied. The first target was the current perceptual decision, that is, whether the participant responded cardinal or diagonal. The second target was

whether the participant changed or repeated the previous decision. This second target was especially relevant for the project because it was directly related to decision stability and confirmation bias.

The input variables were grouped into four categories according to the type of information they represented: sensory evidence, previous-decision history, behavioral variables and metacognitive variables. The sensory variables described the current stimulus, including the decision variable and the values of the individual elements in the sequence. These variables were important because they represented the new evidence available in the current trial. The previous-decision history variables included earlier choices, whether the previous response was correct and previous confidence. These variables were central to the project because they captured the information that could bias later decisions. The behavioral variables included reaction time, repetition number and whether the trial was an exact repetition. Finally, metacognitive efficiency was included to explore whether individual differences in metacognition helped predict later behavior. Metacognitive efficiency refers to how well participants can evaluate the accuracy of their own decisions, independently of their general task performance.

This organization was useful because the groups of variables could be added to the models progressively. First, the models were trained using only sensory evidence. Then, variables related to previous decisions, behavior and metacognition were added step by step. This procedure, called feature ablation, made it possible to test whether each group of variables improved prediction beyond the previous ones. For example, if performance increased after adding previous-decision variables, this suggested that decision history contained useful information for predicting later behavior.

Several classifiers were compared, including logistic regression, support vector machines, random forest and gradient boosting. These models were chosen to compare simpler linear approaches with more flexible non-linear models. Models were evaluated using cross-validation grouped by participant, so that trials from the same participant were not used both for training and testing. This was important because it tested whether the models could generalize to participants that had not been seen during training.

Permutation importance was also used to examine which individual variables contributed most to the predictions. In this analysis, one variable is randomly shuffled, so that its relation with the response is altered. If model performance decreases after shuffling that variable, this means that the model was relying on it to make predictions.

Additional classifier comparisons, permutation importance analyses and individual-difference results are reported in Appendix C.

5.4 EEG Analyses

EEG analyses were used to explore whether brain activity contained information about the visual stimuli, the final decisions and the confidence of the participants. This was important because the behavioral and computational analyses described the responses of the participants and modeled the decision process, whereas EEG allowed us to examine

whether these effects were also reflected in neural activity during the task.

The first step was to compute grand average responses. These responses were obtained by averaging the EEG signal across trials and participants. This analysis was useful for visualizing the general temporal structure of the task and for identifying the main visual responses produced by the stimulus sequence. The grand averages were also used to examine whether visual responses changed across repetitions. This was done mainly in occipital channels, which are located at the back of the head and are especially related to visual processing. Among these channels, Oz was selected as a representative channel because it is positioned over visual areas.

The main EEG analysis was orientation decoding. Decoding analyses test whether information about a variable can be extracted from patterns of brain activity. In this case, the aim was to predict the orientation of each visual stimulus from the EEG signal at different time points. Because stimulus orientation is a circular variable, the main model used for this analysis was AngularRegression with LinearSVR. This model transformed each orientation into sine and cosine components and predicted these components from EEG activity. Performance was evaluated using circular correlation, which measured how close the predicted orientation was to the real orientation.

Orientation decoding was also examined separately for the six stimuli in each sequence and for the three stimulus presentations: $nrep = 0$, $nrep = 1$ and $nrep = 2$. This separation was important because the first presentation did not contain a previous decision, whereas later presentations could already be influenced by decision history. Comparing orientation decoding across repetitions made it possible to test whether the neural representation of stimulus orientation changed when the same orientations were presented again.

Decision decoding was also performed to test whether the final decision of the participant could be predicted from EEG activity. In this analysis, Linear Discriminant Analysis (LDA) was used to classify whether the participant responded cardinal or diagonal at each time point. This analysis was performed separately for $nrep = 0$, $nrep = 1$ and $nrep = 2$. This was important because, in the first presentation, there was no previous decision that could bias the response of the participant. In contrast, in later presentations, the previous decision could influence the neural state before the new stimulus sequence appeared. Therefore, separating the three presentations helped clarify whether decision-related information in the EEG was mainly linked to the current stimulus or whether it could also reflect the influence of previous decisions.

To further examine the role of previous decisions, additional analyses separated the data according to whether the current evidence was congruent or incongruent with the previous response. These analyses were performed only for $nrep = 1$ and $nrep = 2$, because a previous decision was needed to define congruence.

For these analyses, each stimulus was compared with the previous decision of the participant. A stimulus was considered congruent when its orientation supported the same category chosen before and incongruent when it supported the opposite category. For example, after a previous cardi-

nal response, cardinal evidence was congruent and diagonal evidence was incongruent. The same logic was applied at the trial level. A trial was considered congruent when most of its six stimuli supported the previous decision and incongruent when most of them supported the opposite decision.

This separation allowed us to test whether evidence that confirmed or contradicted the previous choice was represented differently in the EEG signal. Orientation decoding was compared between congruent and incongruent stimuli, while decision decoding was compared between congruent and incongruent trials.

Finally, several complementary EEG analyses were performed. LDA-based orientation classification was used as a simpler classification approach to orientation decoding. Transfer decoding from the localizer task to the main task was used to test whether neural patterns learned in a simpler sensory task generalized to the main decision-making task. Some decision decoding methods were also compared, including LDA variants and Multilayer Perceptron Models (MLP). Confidence decoding was performed using Support Vector Regression (SVR) to test whether participants' confidence ratings could be predicted from EEG activity.

The main EEG results focus on the comparison across repetitions and on the congruency analyses, because these analyses are more directly related to confirmation bias. Complementary EEG analyses, including transfer decoding, comparison of decision decoding methods and confidence decoding, are included in Appendix D.

6 RESULTS

The results are presented following the different analyses carried out in the project. First, the behavioral results describe how participants responded across the different presentations, focusing on when they maintained or changed their previous decisions. Then, the computational modeling results examine which part of the decision process was most influenced by previous choices. The machine learning analyses provide a complementary perspective by testing whether later behavior could be predicted from sensory information, decision history and confidence. Finally, the EEG results explore whether these behavioral effects were also reflected in brain activity.

6.1 Behavioral Results

The first analyses focused on the general behavior of the participants during the task. Accuracy and reaction times were stable across stimulus presentations, suggesting that participants performed the task consistently.

The psychometric analyses showed that participants remained sensitive to the current sensory evidence across repetitions (Appendix A, Fig. 10). As sensory evidence became stronger, the probability of responding diagonal increased in the expected direction. This indicates that participants were still taking the current stimulus into account, even though their later responses were also influenced by previous decisions.

The clearest behavioral effect was the tendency to repeat earlier decisions. When participants saw the visual evidence again, they repeated their previous response in approximately 84% of the trials. This happened when the ex-

act same sequence was repeated and when the same orientations were presented again but in a different order. Therefore, this repetition effect was not simply explained by seeing the exact same sequence again. Instead, it suggests that the previous decision itself strongly influenced the later response.

This pattern is shown in Fig. 2, which summarizes the probability of repeating the previous decision across repetitions and trial types. After the first presentation, participants tended to repeat their previous decision and this tendency remained high across the subsequent presentations. This happened not only when the exact same sequence was repeated, but also when the order of the sequence changed while the overall evidence remained similar. For this reason, the effect does not seem to come only from seeing the same stimulus again. Rather, it suggests that the later responses of the participants were influenced by how they had previously interpreted the evidence. The right panel supports this interpretation using the GLMM switch model. Since this model predicted whether participants changed their previous decision, negative estimates indicate a lower probability of switching and therefore a greater tendency to maintain the previous response after repeated presentations.

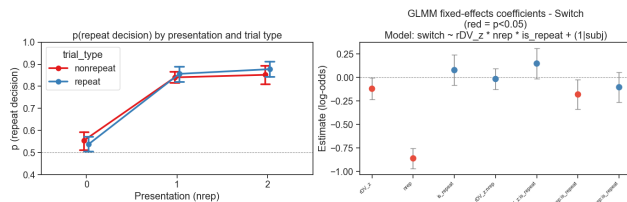


Fig. 2: Decision repetition across stimulus presentations. The left panel shows the probability of repeating the previous decision across repetitions, separately for repeated and non-repeated sequences. Participants showed an increase in decision repetition from the first to later presentations. The right panel shows the GLMM fixed-effect estimates from the switch model, showing that participants were less likely to change their previous decision after repeated presentations.

A further analysis was used to check whether participants were only repeating their previous answer or whether they were still using the new sensory information. For this, the effect of each stimulus in the sequence was analyzed together with the effect of the previous decision. This analysis was important because a high repetition rate alone does not tell us whether participants ignored the new stimulus or whether they integrated it in a biased way.

The results showed that the individual stimuli still influenced the later response, meaning that participants were still using the current visual evidence. However, the previous decision received a much larger estimate than any individual stimulus. This is summarized in Table 1: the estimate for the previous decision was 2.969 in $nrep = 1$ and 3.125 in $nrep = 2$, whereas the individual stimulus estimates remained below 0.44 in both cases. This suggests that participants processed the new information, but their later decisions were much more strongly shaped by what they had decided before.

Presentation	Prev. decision	Indiv. stimuli	p-values
$nrep = 1$	2.969	0.313–0.438	$p < .001$
$nrep = 2$	3.125	0.283–0.434	$p < .001$

TABLE 1: Comparison between the effect of previous decisions and individual stimuli. This table summarizes the model estimates for the previous decision and for the individual stimuli in the sequence. The previous decision had a much larger estimate than any individual stimulus in both repetitions, although the individual stimulus effects were also significant.

The comparison shown in Fig. 3 helps clarify whether later responses were driven only by the previous decision or whether the current stimulus sequence still contributed to the response. The left panel shows the estimated weights of the individual stimuli in the current sequence, ordered according to whether they were congruent or incongruent with the previous decision. Some of these stimulus weights were different from zero, suggesting that current sensory information still influenced participants' responses. However, these weights were smaller and more variable than the weight of the previous decision. As shown in the right panel, the previous decision had a larger positive weight in both repetitions. This indicates that participants did not simply ignore the current stimulus, but their later responses were strongly biased by what they had decided before.

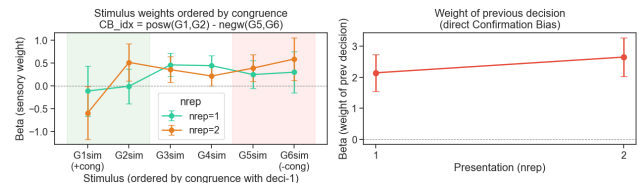


Fig. 3: Influence of current sensory evidence and previous decisions. The left panel shows the estimated weights of the individual stimuli in the current sequence, ordered according to their congruence with the previous decision. The right panel shows the estimated weight of the previous decision across repetitions. The results indicate that participants still used current sensory information, but the previous decision received a larger and more consistent model weight than the individual stimulus estimates.

Confidence also played an important role. Participants were more likely to repeat a previous decision when they had been highly confident in that response, while lower confidence was associated with more changes of mind (Appendix A, Fig. 11). Additional analyses also showed that this effect depended partly on whether the previous decision had been correct or incorrect: repetition was especially high after correct high-confidence decisions and lower after incorrect low-confidence decisions (Appendix A, Fig. 12). This suggests that confidence helped stabilize previous choices across repetitions.

Overall, the behavioral results show that later perceptual decisions were not based only on the current stimulus. Participants continued to use current sensory evidence, but their responses were also strongly influenced by previous decisions and by the confidence associated with those decisions. This pattern is consistent with perceptual confirmation bias.

6.2 Computational Modeling Results

The HSSM analyses were used to better understand how the behavioral bias could emerge during the decision process. The behavioral results showed that participants repeated previous decisions very often, but this did not explain the mechanism behind this effect. For this reason, computational models were used to test whether confirmation bias acted mainly as an initial bias before the new evidence was processed or as a bias during the accumulation of the new evidence.

Two possible mechanisms were compared. In the first model, the previous decision biased the starting point of the decision process. This means that the participant started the next decision already closer to the response chosen before. In the second model, the previous decision biased the drift rate. This means that the previous choice affected how new sensory evidence was accumulated during the decision.

Before fitting the models to the real data, simulations and parameter recovery analyses were carried out. This step was used to check whether the models could detect the bias effects when these were already known. Since the models were able to recover the simulated effects, the modeling approach could be applied to the real experimental data with confidence.

The two models were then compared directly using LOO-CV. This comparison favored the drift rate bias model over the starting point bias model. Although the *elpd_loo* values were negative in both models, what matters for this measure is which value is higher or less negative. In this case, the drift rate model had the higher *elpd_loo* value (-2298.16) compared with the starting point model (-2539.44), indicating better predictive performance. The comparison between the two models is summarized in Fig. 4, which shows both the LOO-CV results and the predictive accuracy of each model.

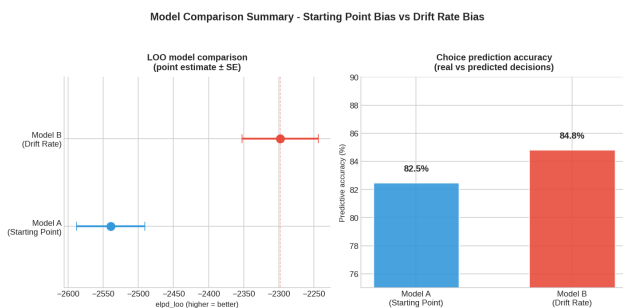


Fig. 4: Model comparison between starting point bias and drift rate bias models. The left panel shows the LOO model comparison between the two HSSM models, where higher *elpd_loo* values indicate better predictive performance. The drift rate bias model showed a higher *elpd_loo* value than the starting point bias model, indicating better model fit. The right panel shows the predictive accuracy of each model when comparing predicted decisions with the real decisions of the participants. The drift rate bias model also showed higher predictive accuracy. Together, these results indicate that the model in which the previous decision biases the drift rate provided the best account of the data.

The model weight also supported this result. Model weight can be understood as the relative support for each model when the two models are compared. The drift rate

model received a much higher weight than the starting point model (94.6% vs. 5.4%), meaning that the data was much more consistent with the drift rate explanation. The full model comparison, including predictive accuracy and *elpd_loo* values, is reported in Appendix B, Table 3.

As an additional check, predictive accuracy was calculated by comparing the decisions predicted by each model with the real decisions made by the participants. The starting point model correctly predicted 82.5% of the real decisions, whereas the drift rate model correctly predicted 84.8%. Although this difference was not very large, it followed the same direction as the LOO-CV comparison and again favored the drift rate model.

The parameter estimates also helped clarify why the drift rate model was the most informative for this project. In this model, the previous decision had a positive effect on the drift rate ($v_bias_cov = 0.772$, 94% CI [0.719, 0.826]). This means that previous choices influenced the accumulation of new sensory evidence during the next decision. The credible interval (CI) shows the range of values that are most compatible with the data and in this case it was clearly above zero. Therefore, this effect can be interpreted as reliable in the model.

Current sensory evidence also had a positive effect on the drift rate ($v_rdv_z = 0.364$, 94% CI [0.305, 0.417]), showing that participants still used the current stimulus. However, in this model, the estimated effect of the previous decision was larger than the effect of the current sensory evidence. The starting point model also estimated a previous-decision bias, but it showed lower predictive performance. Therefore, the main interpretation focused on the drift rate model, while the full parameter estimates for both models are reported in Appendix B, Table 4.

Overall, the HSSM results suggest that the influence of previous decisions was better explained by a bias in evidence accumulation than by a simple shift in the starting point of the decision process. This is important because it indicates that confirmation bias was not only an initial tendency to repeat the previous response. Instead, previous decisions appeared to shape how new sensory evidence was accumulated during the decision itself.

6.3 Machine Learning Results

The machine learning analyses were used to test whether the behavior of the participants could be predicted from the variables available in the task. This analysis was complementary to the behavioral and computational results. Instead of only testing whether a variable had a significant effect, the machine learning models tested whether the combination of several variables was useful for predicting later behavior.

Four classifiers were compared: logistic regression, linear SVM, random forest and gradient boosting. The models were evaluated using GroupKFold cross-validation. In this procedure, each fold trains the model with data from some participants and tests it on different ones. This makes the analysis more demanding, because it tests real generalization to new participants, instead of only predicting new trials from participants already seen during training. The full classifier comparison is reported in Appendix C.1, Fig. 15 and Table 5.

Two targets were analyzed. The first one was switch, which indicated whether participants changed or repeated their previous decision. The second one was deci, which indicated the current perceptual decision, cardinal or diagonal. Model performance was compared with chance level, which is the performance expected if the model had no useful information to classify the response. For both targets, performance was higher than this baseline, although prediction was better for the current decision than for decision change.

For switch, logistic regression and linear SVM reached similar accuracies, 0.677 and 0.678, with AUC values of 0.735 and 0.731. Random forest performed better, with an accuracy of 0.729 and an AUC of 0.745. Gradient boosting showed the highest accuracy, 0.841, but its F1-score was very low. This suggests that it was not classifying both classes in a balanced way. Therefore, accuracy alone was not enough to evaluate model performance. This result also shows that predicting decision change was difficult, probably because changes of mind were less frequent than repetitions.

For deci, prediction performance was higher than for switch. Logistic regression reached an accuracy of 0.784 and an AUC of 0.861, while linear SVM reached an accuracy of 0.763 and an AUC of 0.852. The best overall performance was obtained with random forest, with an accuracy of 0.811, an AUC of 0.885 and an F1-score of 0.826. This shows that the current perceptual decision could be predicted from the available variables.

The feature ablation analysis was used to test which groups of variables were most useful for prediction. The model was first trained using only sensory variables. Then, variables related to previous decisions, behavior and metacognition were added progressively. This made it possible to test whether each group added useful information beyond the previous ones.

For switch, sensory variables alone produced low performance, with an accuracy of 0.538 and an AUC of 0.548. When history and confirmation bias variables were added, performance increased to an accuracy of 0.665 and an AUC of 0.728. Adding behavioral variables further increased performance to an accuracy of 0.685 and an AUC of 0.742. Adding metacognitive efficiency did not improve the model further. This pattern suggests that the tendency to repeat or change a decision was not mainly explained by the current stimulus, but by information about previous decisions and confidence. For deci, sensory variables already predicted the current decision, with an accuracy of 0.724 and an AUC of 0.803. Adding history and confirmation bias variables increased performance to an accuracy of 0.804 and an AUC of 0.878. The final model, including behavioral and metacognitive variables, reached an accuracy of 0.811 and an AUC of 0.885. This shows that current sensory evidence was important for predicting the decision, but previous-decision history also added relevant information.

The results of the feature ablation analysis are shown in Fig. 5, where performance is compared across progressively larger feature sets. For switch, performance was close to chance when only sensory variables were used, but it increased clearly after adding history and confirmation-bias variables. This suggests that predicting whether participants repeated or changed their previous decision depended

mainly on information about previous choices and confidence. For deci, sensory variables already predicted the current decision relatively well, but performance also improved after adding history and confirmation-bias variables. Overall, this analysis shows that current sensory evidence was useful for predicting decisions, but previous-decision history provided additional information, especially for predicting decision stability.

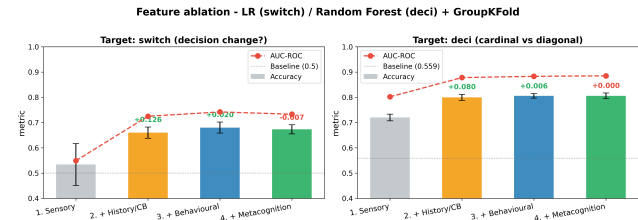


Fig. 5: Feature ablation analysis for machine learning prediction. The figure shows how prediction performance changed when different groups of variables were added to the model. For switch, performance improved after adding history and confirmation-bias variables. For deci, sensory variables already predicted the current decision well, but adding previous-decision history also improved performance. This shows that previous choices provided useful information beyond the current stimulus.

Permutation importance was then used to identify which individual variables contributed most to the predictions. In this analysis, each variable is randomly shuffled and the decrease in model performance is measured. If performance decreases after shuffling a variable, this indicates that the variable contained useful information for the prediction. For switch, the variables producing the largest decrease in performance were related to previous-decision history, especially whether the previous response had been correct and previous confidence. For deci, the most important predictors were the previous decision and the current sensory evidence. This suggests that predicting decision changes depended mainly on previous-trial information, whereas predicting the current decision depended on both the current stimulus and what the participant had decided before. The complete permutation importance results are reported in Appendix C.2, Fig. 16 and Table 6.

Finally, the Leave-One-Subject-Out (LOSO) analysis was used to examine that prediction performance varied between participants. In this analysis, the model is trained on all participants except one and then tested on the participant left out. This makes it possible to examine how well the model generalizes to each individual participant. Mean accuracy was 0.850 for switch and 0.810 for deci, but with considerable variability across participants. This suggests that some participants showed more predictable behavior than others. The individual LOSO results are reported in Appendix C.4, Table 7.

Overall, the machine learning results support the behavioral and computational analyses. They show that previous decisions and confidence were useful predictors of later behavior, especially when predicting whether participants would maintain or change their previous response. This suggests that the tendency to repeat previous decisions was not random, but was related to specific information from the previous trial.

6.4 EEG Results

The EEG analyses were used to examine whether brain activity contained information about the visual stimuli, the final decision and the confidence of the participants. The main results focused on whether these neural representations changed across repetitions and whether they depended on the congruence between the current evidence and the previous decision.

The first step was a descriptive analysis of the grand average responses. The EEG signal showed clear visual responses after stimulus presentation, especially over occipital electrodes. These responses are shown in Appendix D.1, Fig. 18 and Fig. 19.

To describe the general EEG response to the visual stimuli, the Oz channel was used as a representative occipital channel. This channel is located at the back of the head, over visual areas, so it is especially useful for observing visual responses. The response was larger during the first presentation ($nrep = 0$) and became smaller in later repetitions. This suggests that seeing the same orientations again was associated with a weaker visual response. One possible explanation is that the brain responded less strongly because the information had already been presented before.

The main EEG decoding analysis examined whether stimulus orientation could be predicted from EEG activity. Using the orientation decoding approach described in the Methodology section, performance was measured as the circular correlation between the real and predicted orientations. Higher values therefore indicated more accurate orientation decoding.

The orientation decoding analyses separated by repetition are shown in Fig. 6. The results showed that stimulus orientation could be decoded from EEG activity, with increases in decoding appearing around the presentation time of some stimuli. Before comparing repetitions, baseline correction was applied. This means that the decoding level before the stimulus appeared was used as a starting reference and was subtracted from the rest of the signal. In this way, the analysis focused on how much decoding increased or decreased after the stimulus appeared, instead of being influenced by differences that were already present before the stimulus. After separating the analysis by repetition, the clearest decoding increase was observed for the first stimulus of the sequence, O1, where decoding increased around the moment when the stimulus appeared, especially in the first presentation. Smaller increases were also visible for some later stimuli, but the pattern across repetitions was more variable and did not remain consistent across the whole sequence. This suggests that EEG activity contained information about stimulus orientation, although repetition-related changes in orientation decoding were modest and should be interpreted cautiously.

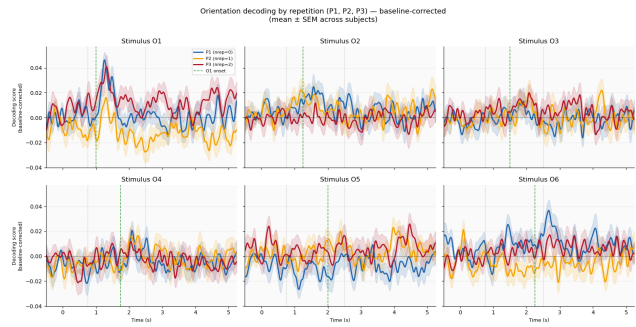


Fig. 6: Orientation decoding from EEG activity across repetitions. The figure shows orientation decoding for each stimulus in the sequence, separated by repetition number. Decoding performance was measured using circular correlation between the real and predicted orientations. The curves were also smoothed to make the general temporal pattern easier to see. Decoding tended to increase around the time when some stimuli appeared, especially O1, suggesting that EEG activity contained information about stimulus orientation. Differences between repetitions were variable, indicating that repetition-related changes in orientation decoding were modest.

Decision decoding was then analyzed separately across repetitions. This analysis was important because the previous versions of the decoding averaged trials from all repetitions together, which made it difficult to interpret whether decision-related information was linked to the current stimulus or to the influence of previous decisions. By separating $nrep = 0$, $nrep = 1$ and $nrep = 2$, it was possible to examine whether decision-related information was already present before the stimulus sequence and whether this changed across repetitions.

As shown in Fig. 7, the results showed that decision decoding was generally above chance, but the pattern differed across repetitions. In the first presentation, decoding accuracy was lower and remained closer to chance, especially before the stimulus appeared. In later presentations, and particularly in $nrep = 2$, decoding accuracy was higher. This suggests that after participants had already seen the same orientations and made a previous decision, EEG activity contained more information about the upcoming decision.

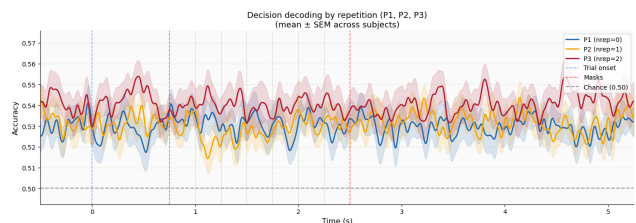


Fig. 7: Decision decoding from EEG activity across repetitions. This figure shows decision decoding accuracy over time, separated by repetition number. The model classified the final response of the participant as cardinal or diagonal, with chance level at 0.50. Curves were smoothed to make the general temporal pattern easier to visualize. Decoding accuracy was closer to chance during the first presentation, but increased in later repetitions, especially in $nrep = 2$. This suggests that decision-related information in EEG activity became stronger across repetitions and may partly reflect the influence of previous decisions.

Importantly, the increase in decision decoding in later repetitions was also visible before the stimulus sequence started. This pattern should be interpreted cautiously, but it is consistent with the idea that previous decisions may influence the neural state before new sensory evidence is presented. Therefore, the decision decoding results suggest that later choices were not only related to the processing of the current stimulus, but also to information carried over from previous decisions.

Additional analyses tested whether EEG decoding changed depending on whether the current evidence matched or contradicted the previous decision. These analyses were only performed for $nrep = 1$ and $nrep = 2$, because a previous decision was needed to classify the evidence as congruent or incongruent.

As shown in Fig. 8, for orientation decoding, incongruent stimuli tended to show stronger decoding than congruent stimuli in some parts of the sequence. This pattern was clearest for O1, where both congruent and incongruent stimuli showed a decoding peak around the presentation of the stimulus, but the peak was higher for incongruent stimuli. A similar tendency appeared for some later stimuli, especially O2 and O5, although it was less consistent across the full sequence. This suggests that evidence contradicting the previous decision may have been represented more strongly in EEG activity than evidence supporting it.

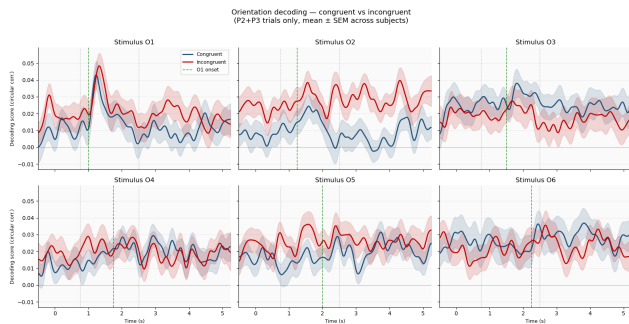


Fig. 8: Orientation decoding for congruent and incongruent stimuli. This figure shows orientation decoding for each stimulus in the sequence, separated according to whether the stimulus matched or contradicted the previous decision. Decoding performance was measured using circular correlation between the real and predicted orientations. Curves were smoothed to make the general temporal pattern easier to visualize. Incongruent stimuli tended to show stronger decoding than congruent stimuli, especially for O1 and some later stimuli. This suggests that evidence contradicting the previous decision may have been represented more strongly in EEG activity, although this result should be interpreted cautiously because the number of trials in each condition was reduced.

Decision decoding was also compared between congruent and incongruent trials. Decoding accuracy was higher for incongruent trials than for congruent trials across much of the trial. This may indicate that trials in which the current evidence conflicted with the previous decision generated a more distinct neural pattern, possibly reflecting greater conflict or increased processing of unexpected evidence. However, this result should also be interpreted cautiously because separating trials according to congruence reduced the amount of data available for each condition. This comple-

mentary result is shown in Appendix D.3, Fig. 21.

Additional EEG analyses, including transfer decoding, comparison of decision decoding methods and confidence decoding, are also reported in Appendix D.4-D.6.

Overall, the EEG results suggest that previous decisions were associated with changes in both sensory and decision-related neural activity. Orientation decoding tended to be weaker on later repetitions, suggesting that the representation of the stimulus orientation may have become less strong once the same orientations had already been presented. In contrast, decision decoding became stronger across repetitions, especially in later presentations. This suggests that, in later repetitions, the EEG signal before the stimulus appeared may already have reflected the influence of the previous decision of the participant. Finally, the congruency analyses suggest that evidence inconsistent with the previous decision may produce stronger neural responses than evidence consistent with it. Taken together, these results provide complementary neural evidence that prior decisions influenced the processing of new sensory information.

7 CONCLUSIONS

This project aimed to understand how previous perceptual decisions influence later decisions when the same visual information is presented again. Overall, the results suggest that later decisions are not based only on the sensory evidence available in the current moment. Instead, they are also shaped by what participants had decided before and by how confident they were in that previous decision.

One of the main ideas that emerges from this project is that decision repetition is not simply a mechanical tendency to give the same response again. Participants repeated their previous decisions very often, but they were still sensitive to the current stimulus. This means that they were not ignoring the new sensory information. Rather, their later decisions seemed to depend on both the new evidence and the interpretation they had already formed in the previous presentation. In this sense, the results support the idea that perceptual confirmation bias can influence how new evidence is used, not only which final response is selected.

Confidence also appears to be important for understanding why some decisions remain stable over time. Participants were more likely to repeat a previous decision when they had reported high confidence in it, especially when that previous decision had been correct. This suggests that confidence may act as a signal that reinforces previous interpretations. When participants trusted their earlier decision, they were more likely to maintain it when the evidence appeared again.

The computational modeling results helped clarify how this bias could arise during the decision process. The HSSM analyses suggested that previous decisions were related not only to an initial bias to the previous response, but also to changes in the accumulation of new sensory evidence. This is important because it suggests that confirmation bias may act during the processing of the new information itself, rather than only at the moment of choosing a response.

The machine learning analyses provided a complementary view of the same pattern. Previous-decision history and

confidence improved the prediction of later behavior, especially when predicting whether participants would repeat or change their previous response. This shows that decision history contained useful information about future choices and that the tendency to repeat previous decisions was systematic, not random.

The EEG results provided a complementary neural perspective. Overall, the analyses showed that stimulus orientation could be decoded from EEG activity, suggesting that the brain signal contained information about the sensory evidence presented in the current trial. When orientation decoding was analyzed separately across repetitions, the effect was still present, but the differences between repetitions were modest and became more variable after baseline correction.

Decision decoding showed a slightly different pattern. Decoding tended to increase across repetitions and in later presentations some decision-related information was already visible before the new stimulus sequence started. This pattern should be interpreted cautiously, but it suggests that previous decisions may influence the neural state in which new evidence is processed. Congruency analyses also suggested that evidence inconsistent with the previous decision may produce stronger neural responses than evidence consistent with it.

There are also some limitations that should be considered. First, although the behavioral and computational results were quite consistent, the EEG analyses were more exploratory. Some EEG effects were modest and separating the data by repetition and congruence reduced the number of trials in each condition. This makes the neural results noisier and limits how strongly they can be interpreted. Second, the computational models provide useful explanations of the decision process, but they are still models. They help identify some mechanisms, but they do not directly prove the exact processes taking place in the brain. Finally, this project used a specific perceptual decision-making task, so future studies would be needed to test whether the same pattern appears in other perceptual tasks or in more realistic situations.

Future work could extend these findings in several directions. One important next step would be to extend the EEG analyses by applying more formal statistical tests to determine which decoding effects are robust and which may reflect noise. For example, cluster-based permutation tests could be used to evaluate whether patterns that appear across nearby time points are stronger than would be expected by chance. Another interesting direction would be to examine individual differences more directly by testing whether participants who show a stronger behavioral confirmation bias also show stronger computational or neural effects, such as larger bias parameters in the HSSM models or stronger EEG decision-related signals. Finally, future studies could benefit from including more participants or a larger number of trials, especially for the EEG and machine learning analyses, in order to obtain more stable estimates and to better evaluate more complex predictive models.

In conclusion, this project suggests that perceptual confirmation bias is not only a tendency to repeat a previous response. Previous decisions and confidence seem to shape how new sensory evidence is used when similar information is presented again. By combining behavioral analyses,

computational modeling, machine learning and EEG, this work shows how previous interpretations can remain stable over time and why changing an initial perceptual decision may be difficult, even when new evidence is available.

8 ETHICAL, SOCIAL AND LEGAL CONSIDERATIONS

This project uses behavioral and EEG data from human participants, so ethical and legal aspects must be considered. The data should be treated as sensitive information because EEG recordings and behavioral responses can contain information about cognitive processes. For this reason, the privacy of the participants, informed consent, anonymization and secure data handling are essential throughout the project.

From a social and human perspective, the project studies how previous decisions and confidence influence later choices. Although the analyses were conducted in an experimental perceptual task, similar methods could be applied in broader contexts involving human decision-making. Therefore, the results should be interpreted carefully and should not be used to make strong conclusions about individuals outside the experimental setting.

Overall, the project highlights the importance of using data responsibly, respecting privacy, transparency and the limits of interpretation.

9 CODE AVAILABILITY

The project repository is available on GitHub: <https://github.com/martinanapoli/Computational-Neuroscience-Martina-Napoli.git>

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APPENDICES

The appendices include additional analyses, figures and technical details that support the results presented in the main report. These materials are not essential for following the main argument of the project, but they provide more complete information about the analyses performed and the robustness of the results.

A SUPPLEMENTARY BEHAVIORAL ANALYSES

This appendix includes additional behavioral analyses that support the main behavioral results described in the report. These analyses provide more detail about the performance of the participants, reaction times, confidence and psychometric patterns across repetitions.

The first analyses focused on the overall task performance. Reaction times were analyzed across stimulus presentations to assess whether participants performed the task consistently. Additional psychometric analyses were also included to show how the responses of the participants changed depending on how clear the sensory evidence was. These analyses confirmed that participants were still sensitive to the current stimulus, even though their later decisions were influenced by previous choices.

Finally, confidence analyses examined how confidence in a previous decision affected the probability of repeating that decision. These analyses support the main finding that high-confidence decisions were more stable and less likely to be changed in later presentations.

A.1 Reaction Times Across Repetitions

Reaction times were examined across repetitions and trial type to check whether participants became faster or slower when the same or similar evidence was presented again. This analysis was important because the main result of the project was the tendency to repeat previous decisions, so it was necessary to verify that this effect was not simply explained by changes in response speed.

To test this, reaction times were analyzed using an LMM which allowed the effect of repetition and trial type to be examined while taking differences between participants into account. In this model, coefficients close to zero indicate that the effect of a variable was very small or absent. The results showed that reaction times remained relatively stable across repetitions, suggesting that participants were not simply becoming faster or slower during the task.

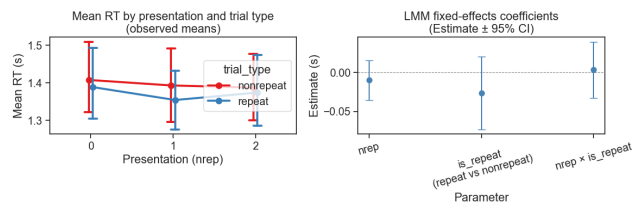


Fig. 9: Reaction times across repetitions and trial type. This figure shows the mean reaction times across the three stimulus presentations, separated by trial type. Reaction times remained relatively stable across repetitions and the LMM coefficients did not show a clear effect of repetition or trial type. This suggests that the main repetition effects observed in the task were not simply due to participants becoming faster or slower across presentations.

A.2 Psychometric Analyses

Psychometric curves were included to examine whether participants were still using the current sensory evidence across repetitions. These curves show the relation between the evidence presented in the stimulus and the response given by the participant. In this case, the x-axis represents the sensory evidence, from more cardinal evidence to more diagonal evidence and the y-axis represents the probability of responding “diagonal”.

If participants were using the current stimulus, the curves should increase from left to right. This means that, as the stimulus contained more evidence for a diagonal response, participants should respond “diagonal” more often. This is what can be observed in Fig. 10. The curves increased with sensory evidence in all repetitions. Therefore, participants remained sensitive to the current stimulus, even in later presentations.

The other panels of Fig. 10 show the same pattern using the model estimates. In the middle panel, the effect of the current sensory evidence is clearly above zero, which means that the stimulus had an important role in the responses. The right panel shows that this effect did not change much across repetitions, because the estimates are closer to zero and the intervals are wider. Therefore, participants seemed to keep using the current stimulus in a similar way across presentations, even though their later decisions were also affected by what they had chosen before.

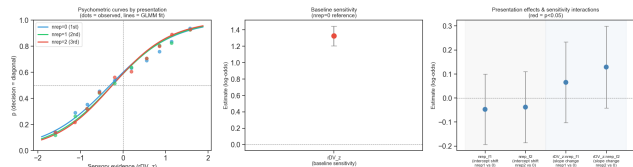


Fig. 10: Psychometric curves across repetitions. The left panel shows the probability of responding “diagonal” as a function of the sensory evidence, separately for each repetition. The increasing curves indicate that participants were more likely to respond “diagonal” when the stimulus provided more evidence for that response. The middle panel shows the baseline sensitivity to sensory evidence and the right panel shows how this sensitivity changed across repetitions. Overall, the results indicate that participants remained sensitive to the current stimulus across repetitions.

A.3 Confidence and Decision Stability

Additional confidence analyses were included to examine whether confidence in a previous decision affected the probability of repeating it later. The relationship between previous confidence and decision repetition is shown in Fig. 11.

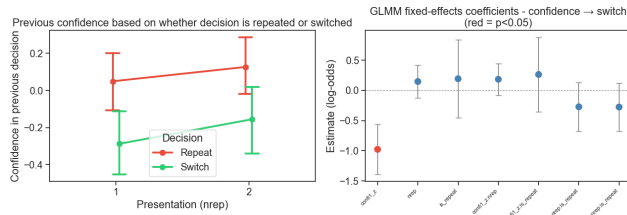


Fig. 11: Confidence and decision repetition. This figure shows how confidence in the previous decision was related to repeating that same decision later. When participants were more confident, they tended to repeat their previous decision more often. However, when confidence was lower, they were more likely to change their mind. This suggests that confidence may have helped stabilize decisions across repetitions.

A.4 Previous Confidence, Previous Accuracy and Decision Repetition

This analysis examined whether the effect of confidence on decision repetition depended on whether the previous decision had been correct or incorrect. This was useful because confidence may not have the same meaning after correct and incorrect decisions. For example, a high-confidence correct decision could make participants more likely to repeat their response, while a low-confidence or incorrect decision could make them more likely to change it. This interaction between previous confidence and previous accuracy is shown in Fig. 12.

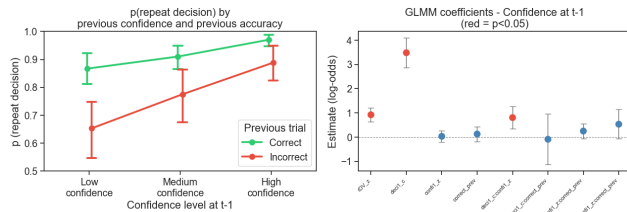


Fig. 12: Previous confidence, previous accuracy and decision repetition. This figure shows how the probability of repeating a previous decision changed depending on confidence and whether that decision had been correct. Participants were more likely to repeat their previous response when they had been highly confident, especially after correct decisions. In contrast, repetition was lower when confidence was low, particularly after incorrect decisions. This suggests that confidence helped make previous perceptual decisions more stable.

B HSSM SIMULATIONS AND MODEL VALIDATION

This appendix includes the additional analyses used to check that the computational models worked properly before applying them to the real data. This step was important

because the models needed to be able to recover known effects in simulated data before being used to interpret the experimental results.

The first part of this appendix presents the simulations of the two possible confirmation bias mechanisms. In the starting point bias model, the previous decision shifted the initial point of the next decision process. In the drift rate bias model, the previous decision changed how new evidence was accumulated. Comparing these simulations helped show what kind of behavioral patterns each mechanism should produce.

The appendix also includes the parameter recovery analyses. In these analyses, data was simulated using known parameter values and then fitted again with the model. If the model recovered the original values correctly, this supported the idea that the model could be applied to the real data with more confidence.

Finally, this appendix includes the main parameter estimates obtained from the real data. These values show how strongly each factor contributed to the decision process in the HSSM models, for example the effect of the previous decision or the effect of the current sensory evidence. They provide a more detailed numerical summary of the modeling results described in the main text.

B.1 Simulated Model Behavior

First, two possible mechanisms of confirmation bias were simulated. In the starting point bias model, the previous decision changed the initial point of the next decision process. In the drift rate bias model, the previous decision changed how new evidence was accumulated during the decision. The simulated psychometric patterns for both mechanisms are shown in Fig. 13.

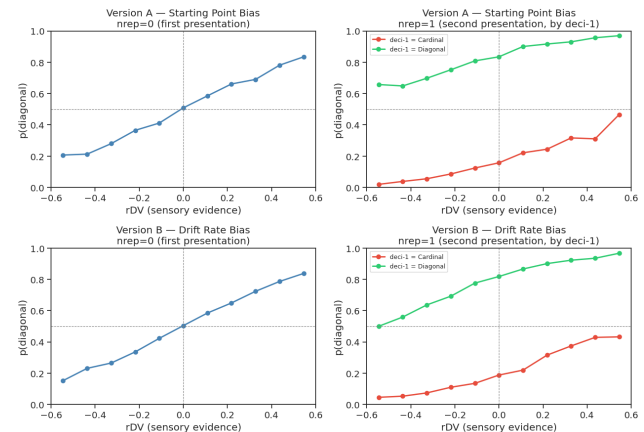


Fig. 13: Simulated psychometric curves for the starting point and drift rate bias models. The figure shows the predicted response patterns for both models. In $nrep = 0$, there is no previous decision, so both models show a single psychometric curve. In $nrep = 1$, the curves are separated according to the previous decision, showing how each model produces confirmation bias.

The two models also produced different reaction time patterns, which helped distinguish between them. The corresponding simulated reaction time distributions are shown in Fig. 14.

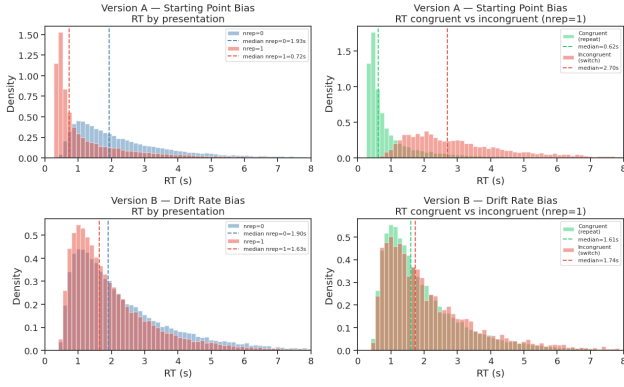


Fig. 14: Simulated reaction time distributions for the starting point and drift rate bias models. The starting point bias model produced longer reaction times for incongruent responses than for congruent responses, whereas this congruency effect was less pronounced in the drift rate bias model. This pattern was useful for comparing the simulated mechanisms with the real behavioral data.

B.2 Parameter Recovery

Parameter recovery was performed to check whether the models could recover known bias effects from simulated data. This step was important because a model should be able to identify a simulated effect before it is used to estimate effects in real data. The parameter recovery results for the two simulated models are summarized in Table 2. In both cases, the estimated bias parameter was positive and its 94% credible interval did not include zero, indicating that the models recovered the simulated bias effect. The numerical values of the starting point and drift rate parameters should not be directly compared with each other, because they are expressed on different model scales.

Model	Parameter	Est. mean	94% CI
Starting point bias	z_z_cov	1.689	[1.602, 1.779]
Drift rate bias	v_v_cov	0.490	[0.437, 0.545]

TABLE 2: Parameter recovery results for the two HSSM models. The models were fitted to simulated data generated with known bias effects. For both models, the recovered bias parameter was positive and the 94% credible interval did not include zero, indicating that the simulated bias was detected by the model. The drift rate parameter is expressed in the scale of the evidence accumulation rate, whereas the starting point parameter is expressed in the internal model scale used to estimate shifts in the initial decision point. For this reason, their numerical values should not be directly compared.

B.3 Model Comparison in the Real Data

After model validation, the two HSSM models were compared using the real experimental data. The aim of this comparison was to determine whether confirmation bias was better explained as a shift in the starting point of the decision process or as a bias in the drift rate.

Model comparison was performed using LOO-CV. This method estimates how well each model predicts unseen data while also taking model complexity into account. Higher

$elpd_{loo}$ values indicate better predictive performance. Predictive accuracy was also calculated as a complementary measure by comparing the decisions predicted by each model with the real decisions made by the participants. This is shown in Table 3.

Model	$elpd_{loo}$	p_{loo}	Model weight	Predictive accuracy
Model A: Starting point bias	-2539.44	10.43	5.4%	82.5%
Model B: Drift rate bias	-2298.16	7.34	94.6%	84.8%

TABLE 3: Model comparison between the starting point bias and drift rate bias models. The drift rate bias model showed a higher $elpd_{loo}$ value, a higher model weight and higher predictive accuracy than the starting point bias model. This indicates that the model in which the previous decision biased the drift rate provided a better account of the data.

The comparison favored the drift rate bias model. This suggests that, in this task, previous decisions influenced later choices mainly by changing how new evidence was accumulated, rather than only by shifting the initial point of the decision process.

B.4 HSSM Estimates in the Real Data

After model validation, the HSSM models were applied to the real experimental data. The main parameter estimates and their 94% credible intervals are summarized in Table 4.

Model	Parameter	Mean	94% credible interval
Model A: Starting point bias	z_bias_cov	0.416	[0.361, 0.472]
Model A: Starting point bias	$z_intercept$	-0.084	[-0.139, -0.027]
Model B: Drift rate bias	v_bias_cov	0.772	[0.719, 0.826]
Model B: Drift rate bias	v_rdv_z	0.364	[0.305, 0.417]
Model B: Drift rate bias	$v_intercept$	0.107	[0.046, 0.173]

TABLE 4: HSSM parameter estimates from the real data. This table summarizes the main parameter estimates and their 94% credible intervals for the HSSM models.

C SUPPLEMENTARY MACHINE LEARNING ANALYSES

This appendix includes additional machine learning results that support the main results of the report. The main text focuses on the most important findings, while this appendix gives a more complete view of the prediction analyses.

First, this appendix includes the full comparison of the classifiers used in the analysis: logistic regression, linear SVM, random forest and gradient boosting. These results show how well each model predicted the two targets: the current decision and whether participants repeated or changed their previous decision.

The feature ablation analysis is shown in the main report, because it is one of the most important machine learning results. In this appendix, the additional analyses focus on classifier comparison, permutation importance, prediction across repetitions and individual differences. These analyses help show which models performed best, which variables were most useful and whether prediction performance changed across participants.

Finally, the appendix includes the permutation importance results and the LOSO analysis. The permutation importance analysis shows which individual variables were

most useful for the models. The LOSO analysis shows how prediction performance changed across participants.

C.1 Classifier Comparison

Several classifiers were compared for the two prediction targets: switch, which indicated whether the participant changed or repeated the previous decision, and deci, which indicated the current perceptual decision. The classifier comparison is shown in Fig. 15 and the corresponding numerical values are summarized in Table 5. Accuracy reflects the overall proportion of correct predictions, AUC-ROC reflects how well the model separates the two classes and F1-score is especially useful when one class is less frequent than the other.

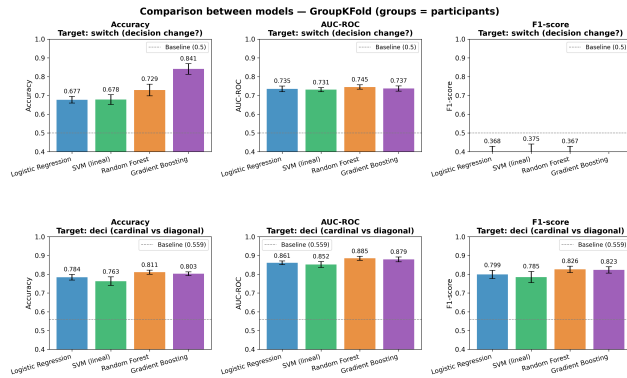


Fig. 15: Classifier comparison for switch and deci. This figure compares the performance of logistic regression, linear SVM, random forest and gradient boosting using accuracy, AUC-ROC and F1-score. For deci, all models showed prediction performance above chance. For switch, performance was also above chance, although F1-scores were lower, showing that decision change was harder to classify in a balanced way.

Target	Model	Accuracy	AUC-ROC	F1-score
switch	Logistic Regression	0.677 ± 0.018	0.735 ± 0.014	0.368
switch	Linear SVM	0.678 ± 0.027	0.731 ± 0.011	0.375
switch	Random Forest	0.729 ± 0.030	0.745 ± 0.012	0.367
switch	Gradient Boosting	0.841 ± 0.027	0.737 ± 0.013	0.132
deci	Logistic Regression	0.784 ± 0.015	0.861 ± 0.010	0.799
deci	Linear SVM	0.763 ± 0.024	0.852 ± 0.015	0.785
deci	Random Forest	0.811 ± 0.011	0.885 ± 0.011	0.826
deci	Gradient Boosting	0.803 ± 0.009	0.879 ± 0.013	0.823

TABLE 5: Accuracy, AUC-ROC and F1-score for all classifiers. This table summarizes the numerical performance of each model for both prediction targets.

C.2 Variable Importance

Permutation importance was used to examine which individual variables contributed most to the predictions. In this analysis, each variable is shuffled and the decrease in model performance is measured. The permutation importance results for individual variables are shown in Fig. 16 and the grouped importance values are summarized in Table 6.

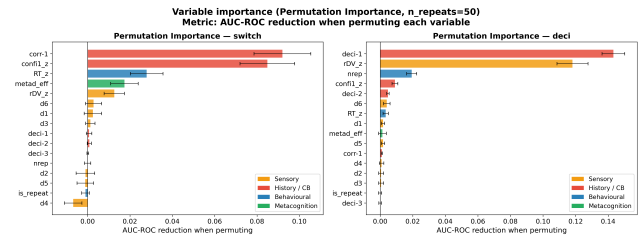


Fig. 16: Permutation importance for individual variables. For switch, the most important variables were related to previous history, especially whether the previous response had been correct and previous confidence. For deci, the most important variables were the previous decision and the current sensory evidence.

Target	Sensory	History / CB	Behavioural	Metacognition
switch	0.0102	0.1783	0.0264	0.0172
deci	0.1278	0.1577	0.0224	0.0016

TABLE 6: Grouped feature importance by variable type. This table summarizes the total importance of each feature group. History and confirmation bias variables contributed most to both prediction targets, especially for predicting decision changes.

C.3 Prediction Across Repetitions

Prediction performance was also examined across repetitions to test whether the models performed differently when more previous information was available. These repetition-specific prediction results are shown in Fig. 17.

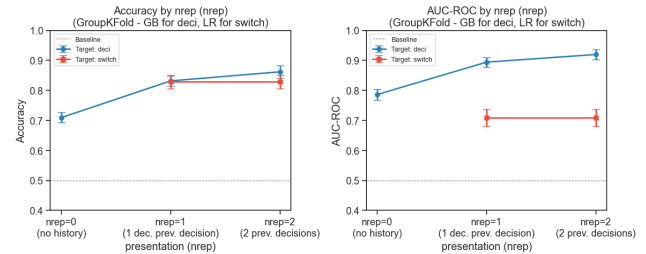


Fig. 17: Prediction performance across repetitions. Prediction of the current decision improved across repetitions, especially in terms of AUC-ROC. This suggests that previous-decisions variables contributed more when the same or similar evidence was presented again.

C.4 Individual Differences

Additional analyses were carried out to examine whether prediction performance differed between participants and whether individual prediction accuracy was related to behavioral measures. This is shown in Table 7.

Analysis	Target	Result
LOSO accuracy	switch	Mean Acc = 0.850, SD = 0.074
LOSO accuracy	deci	Mean Acc = 0.810, SD = 0.046
Correlation with p(maintain decision)	switch	$r = 0.984$, $p < 0.001$
Correlation with metacognitive efficiency	switch	$r = 0.122$, $p = 0.572$
Correlation with mean confidence	switch	$r = 0.231$, $p = 0.277$
Correlation with p(maintain decision)	deci	$r = 0.875$, $p < 0.001$
Correlation with metacognitive efficiency	deci	$r = 0.073$, $p = 0.733$
Correlation with mean confidence	deci	$r = 0.269$, $p = 0.204$

TABLE 7: LOSO summary and individual-difference correlations. This table summarizes individual-level prediction performance and its relation with behavioral measures. The strongest association was found between prediction accuracy and the probability of maintaining the previous decision.

D SUPPLEMENTARY EEG ANALYSES

This appendix includes additional EEG analyses that support the results presented in the main text. The main report focuses on the analyses most closely related to confirmation bias: orientation decoding across repetitions, decision decoding across repetitions and decoding based on congruence with the previous decision. The analyses included here provide extra information about the EEG signal.

D.1 Grand Average Responses

The first analyses show the general EEG response during the main task. They were used to visualize how the EEG signal changed over time and to check whether clear responses appeared after the visual stimuli were presented. The grand average response is shown in Fig. 18 and the Oz response separated by repetition is shown in Fig. 19.

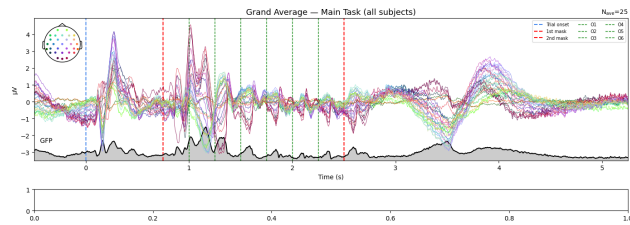


Fig. 18: Grand average EEG responses in the main task. This figure shows the average EEG response across participants during the main task. The vertical lines mark the main events of the trial, including the masks and the six visual stimuli.

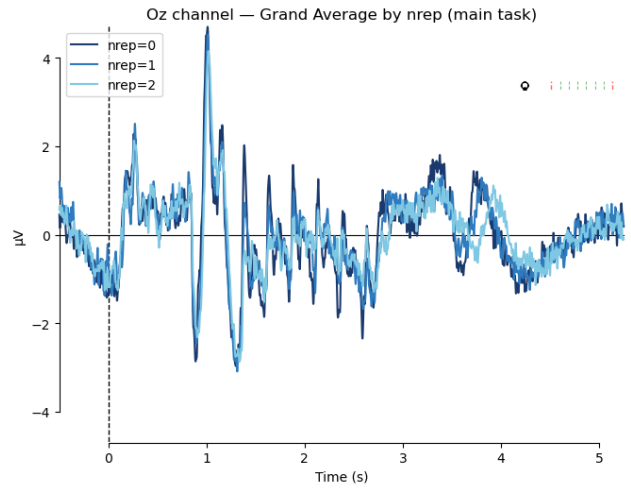


Fig. 19: Oz response separated by repetition. This figure shows the grand average response in the Oz channel for $nrep = 0$, $nrep = 1$ and $nrep = 2$. The response decreased across repetitions, suggesting that repeated exposure to the same orientations was associated with a weaker visual response.

D.2 General Orientation Decoding in the Main Task

Before separating the analyses by repetition, orientation decoding was examined across the full main task. This analysis, shown in Fig. 20, tested whether the orientation of the stimuli could be predicted from EEG activity when all repetitions were considered together. It was useful as a general check that the EEG signal contained information about stimulus orientation during the task.

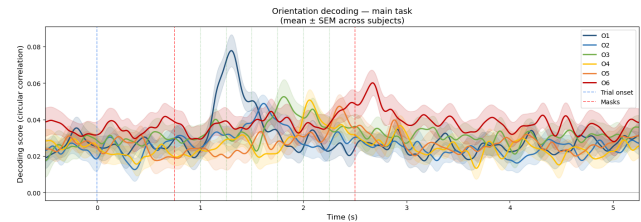


Fig. 20: Orientation decoding in the main task. This figure shows orientation decoding across the full main task, averaging across repetitions. Decoding performance was measured using circular correlation between the real and predicted orientations. The six curves correspond to the six stimuli in the sequence. Curves were smoothed to improve visualization. Decoding peaks appeared around the presentation time of the corresponding stimuli, especially for O1, suggesting that EEG activity contained information about stimulus orientation during the task.

D.3 Congruency-Based EEG Decoding Analyses

Additional analyses examined whether EEG decoding differed depending on whether the current evidence was congruent or incongruent with the previous decision. These analyses were performed only for $nrep = 1$ and $nrep = 2$, because a previous decision was required to define congruence.

For stimulus-level orientation decoding, each stimulus was classified as congruent or incongruent depending on

whether its orientation was consistent with the previous response of the participant. For trial-level decision decoding, each trial was classified as congruent or incongruent depending on whether most of the six stimuli were consistent or inconsistent with the previous decision.

The orientation decoding comparison between congruent and incongruent stimuli is described in the main text, because it was directly related to the question of whether evidence supporting or contradicting the previous decision was represented differently in EEG activity. The corresponding decision decoding comparison is shown in Fig. 21. Decision decoding was higher for incongruent than for congruent trials, suggesting that evidence conflicting with the previous decision may have produced a more distinct neural pattern. However, this result should be interpreted cautiously, because dividing trials by congruence reduced the number of trials available in each condition.

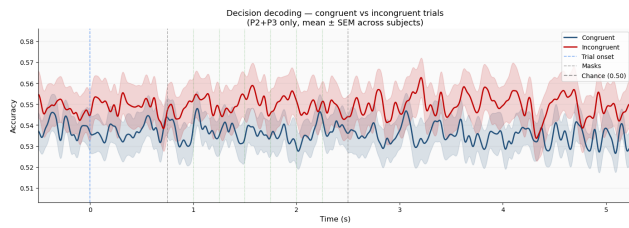


Fig. 21: Decision decoding for congruent and incongruent trials. The figure shows decision decoding accuracy across time for trials classified as congruent or incongruent with the previous decision. Analyses were performed only for $nrep = 1$ and $nrep = 2$.

D.4 Transfer Decoding from the Localizer to the Main Task

Transfer decoding was used to compare the localizer with the main task. The classifier was trained on the localizer and tested on the main task, in order to examine whether the sensory pattern learned in the localizer generalized to the decision-making task. As shown in Fig. 22, transfer decoding performance was lower than orientation decoding in the main task. This suggests that the neural patterns related to orientation in the localizer did not fully generalize to the main task, probably because the main task involved additional decision-related and contextual processes.

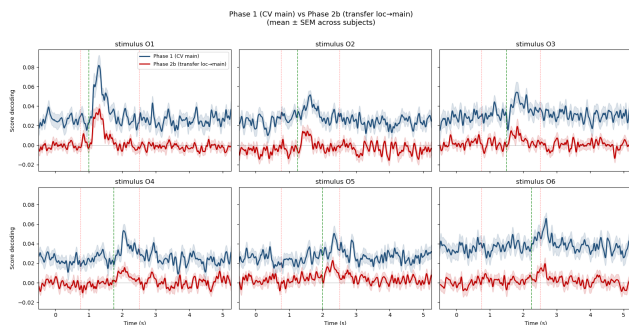


Fig. 22: Transfer decoding from the localizer to the main task. Transfer decoding showed lower performance, suggesting that the neural representation of orientation in the main task was not identical to the simpler sensory response measured in the localizer.

D.5 Comparison of Decision Decoding Methods

Several decision decoding methods were compared to check which type of EEG information was most useful for predicting the final decision of the participant. All these analyses had the same goal: to classify whether the participant responded cardinal or diagonal from the EEG signal.

The comparison included LDA using all channels, which used the full EEG signal. LDA using central channels focused only on electrodes located over central head areas. LDA with spectral features used frequency-based information from the EEG signal. LDA with shrinkage applied a more regularized version of LDA to make the model more stable. LDA with PCA first reduced the dimensionality of the EEG data before classification. Finally, MLP models were included as more flexible neural network models.

This comparison, shown in Fig. 23, helped test whether the decision could be predicted better with simple linear methods or with more complex models.

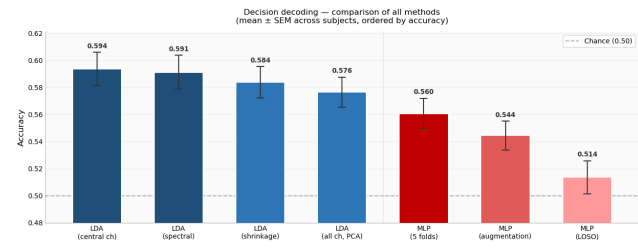


Fig. 23: Comparison of decision decoding methods. This figure compares different methods for predicting the final decision from EEG activity. The LDA-based models performed above chance and showed relatively similar results, while the MLP models performed worse, especially in LOSO validation. This suggests that, for this dataset, simpler linear models were more effective than more complex neural network models.

D.6 Confidence Decoding

The final EEG analysis examined whether confidence could be predicted from EEG. As shown in Fig. 24, confidence was predicted using SVR and performance was evaluated using Spearman correlation between real and predicted confidence ratings.

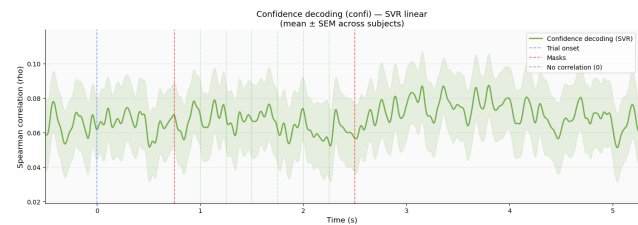


Fig. 24: Confidence decoding from EEG activity across the trial. This figure shows the prediction of confidence from EEG activity. The correlation remained close to zero but positive across the trial, suggesting that EEG activity contained weak information about confidence.